Online Shopping Clickstream & Customer Behavior Analysis using Python

Exploratory Data Analysis (EDA) Report

Dataset: UCI Clickstream Data for Online Shopping (e-shop clothing 2008, April - August 2008)

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

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Project 04 of the Bleep EDA Project Assignment

1. Introduction and Objective

An e-commerce company wants to understand how users interact with its website and to find patterns in online shopping behaviour. In this project I use Python to clean and explore a clickstream log from an online clothing store, and I turn the findings into practical recommendations.

The analysis tries to answer these nine questions:

- Which product categories receive the most attention?
- Which pages attract the most clicks?
- Does product price affect user behaviour?
- Which countries generate the most activity?
- What are the most frequently visited page locations?
- Are there differences in behaviour across product categories?
- What patterns can be identified between browsing activity and purchasing behaviour?
- Which variables appear to be associated with customer conversion?
- Are there unusual user behaviours or outliers?

Important note: the dataset has no purchase column, so real purchases and real conversion cannot be measured. For questions 7 and 8 I therefore use a **proxy**: a 'highly engaged session' is a session with more clicks than 75% of all sessions. This shows serious interest, but it is not a real purchase, and I say so every time I use it.

2. Dataset and Data Understanding

The dataset is the **UCI Clickstream Data for Online Shopping** (https://archive.ics.uci.edu/dataset/553/clickstream+data+for+online+shopping). It records clicks on an online clothing store between 1 April and 13 August 2008. The file has **165,474 rows and 14 columns**. Each row is **one click on one product**, and rows with the same session ID form one visit of one user. So I grouped the rows by session whenever I needed to study behaviour during a visit.

Column	Meaning
year, month, day	Date of the click (April to August 2008)
order	Sequence number of the click inside the session (1 = first click)
country	Country of the IP address (coded 1-47)
session ID	Id of the session (one visit)
page 1 (main category)	1 trousers, 2 skirts, 3 blouses, 4 sale
page 2 (clothing model)	Product code (217 different products)
colour	Colour of the product (coded 1-14)
location	Position of the photo on the page: top left, top middle, top right, bottom left, bottom middle, bottom right
model photography	1 = photo from the front (en face), 2 = profile
price	Price in US dollars
price 2	1 = price is higher than the category average, 2 = not higher

Column	Meaning
page	Page number inside the shop (1 to 5)

3. Data Cleaning

The file uses a semicolon as separator, so it was loaded with sep=';'. I checked the data quality and made the following decisions.

Check	Result	Decision
Missing values	None	Nothing to fill or remove.
Duplicate rows	None	Nothing to remove.
Click order inside sessions	All 24,026 sessions have a proper 1, 2, 3 ... sequence	No change needed.
One country per session	Every session has exactly one country	No change needed.
Product code letter vs category (A trousers, B skirts, C blouses, P sale)	1 row did not match (product A18 listed under skirts)	Removed this single row.
Column price 2 vs price	In every category the products marked above average are really more expensive	Column is consistent, kept as it is.
Codes instead of names	Country, category, colour, location and photo are numbers	Replaced with names from the data description file.

After cleaning, the data has **165,473 clicks** in **24,026 sessions**, on **217 products**, from **47 country codes**. I created new columns: a real date, weekday, and name columns for the coded variables. I also built a session-level table with clicks, deepest page reached, number of different products and categories, average price, share of clicks on above-average-price products, and the country and date of each session.

Outlier analysis (Question 9)

The median session has only 4 clicks and the average is 6.89, so a few very long sessions pull the average up. By the IQR rule, sessions with more than 17 clicks are outliers: this is 1,962 sessions. The 99th percentile is 42 clicks, 141 sessions have 50 or more clicks and 17 sessions have more than 100 clicks. The longest session has **195 clicks on 168 different products** (from Poland on 3 August 2008). I did not delete these sessions, because I cannot prove that they are mistakes: they could be very interested shoppers, but some could also be robots. In the charts I cut the values at the 99th percentile when needed, or used log scales, so that they do not hide normal behaviour.

4. Exploratory Data Analysis

4.1 Product categories (Questions 1 and 6)

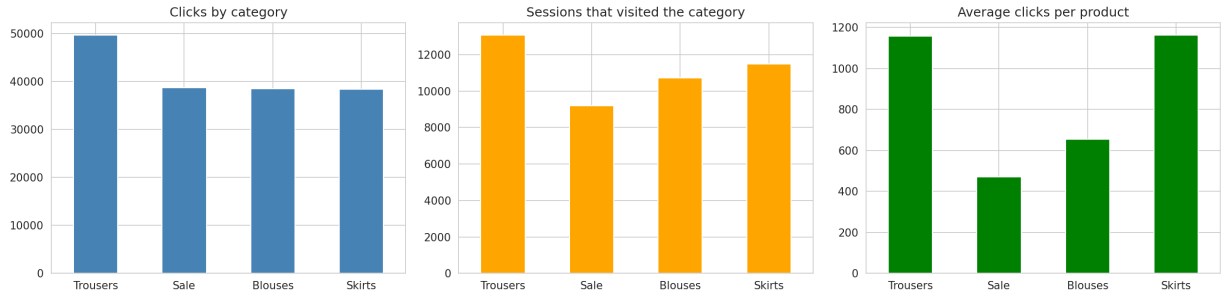


Figure 1: Clicks by category (left), sessions that visited each category (middle) and average clicks per product (right).

What this shows: Trousers get the most attention: 49,742 clicks (30% of all clicks) and 13,082 sessions, which is 54% of all sessions. The other three categories have about the same number of clicks (38,400 to 38,700). But the clicks per product are very different. Skirts (33 products) get **1,164 clicks per product** and trousers (43 products) get 1,157, while blouses (59 products) get 654 and sale (82 products) only **473**. So the sale section has the most products but each product gets the least attention.

Category	Clicks	Products	Average price (USD)	% of clicks on above-average-price items	Average page
Trousers	49,742	43	46.71	34.2%	1.37
Sale	38,747	82	36.23	54.2%	2.19
Blouses	38,577	59	40.29	57.1%	1.99
Skirts	38,407	33	51.19	64.2%	1.39

Skirts are the most expensive category (average price USD 51) and sale is the cheapest (USD 36). Trousers and skirts are mostly shown on the first pages of the shop (average page 1.4), while sale and blouses reach deeper pages (about 2.0 to 2.2).

4.2 Pages and photo locations (Questions 2 and 5)

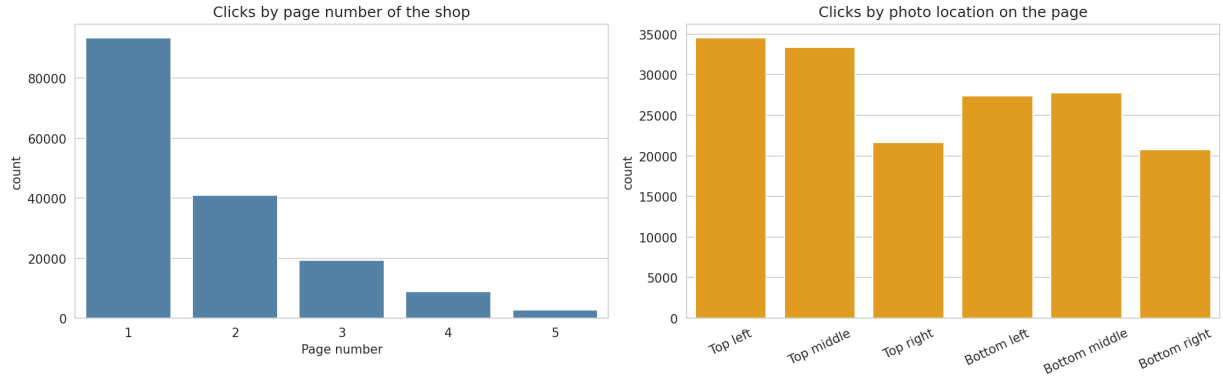


Figure 2: Clicks by shop page number (left) and by photo location on the page (right).

What this shows: Page 1 gets 56.5% of all clicks, page 2 gets 24.8%, page 3 gets 11.7%, page 4 gets 5.4% and page 5 only 1.7%. The number of clicks falls fast with each page, so users rarely go deep into the shop. On the page itself, the **top left (20.9%) and top middle (20.2%)** positions are clicked the most, followed by bottom middle (16.8%) and bottom left (16.5%). The right side is

weaker: top right has 13.1% and bottom right only 12.5%.

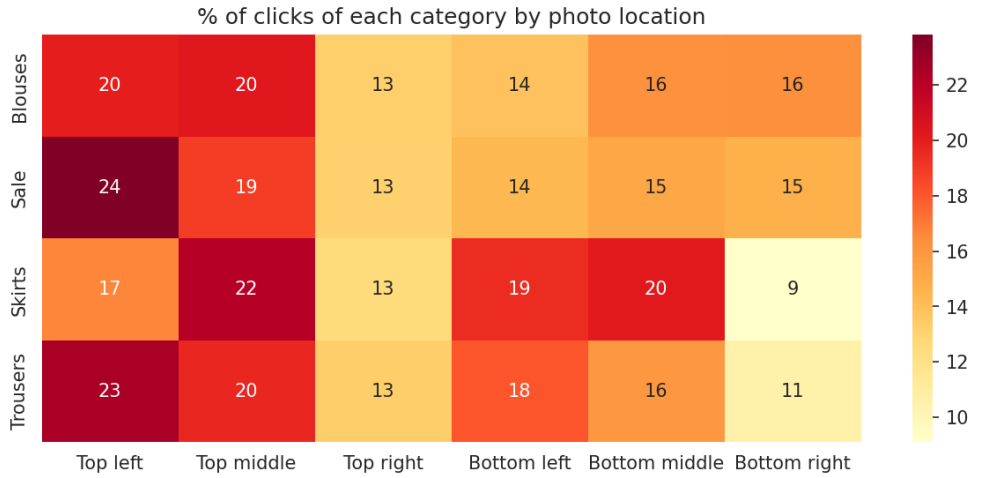


Figure 3: Share of clicks of each category by photo location (%).

What this shows: The pattern is similar for all categories, but with differences. Sale and trousers get most of their clicks in the top left (23.8% and 22.5%), skirts get most in the top middle (22.1%) and the bottom right is weak for skirts (9.1%) and trousers (10.6%). The top right is about 13% for every category.

4.3 Countries (Question 4)

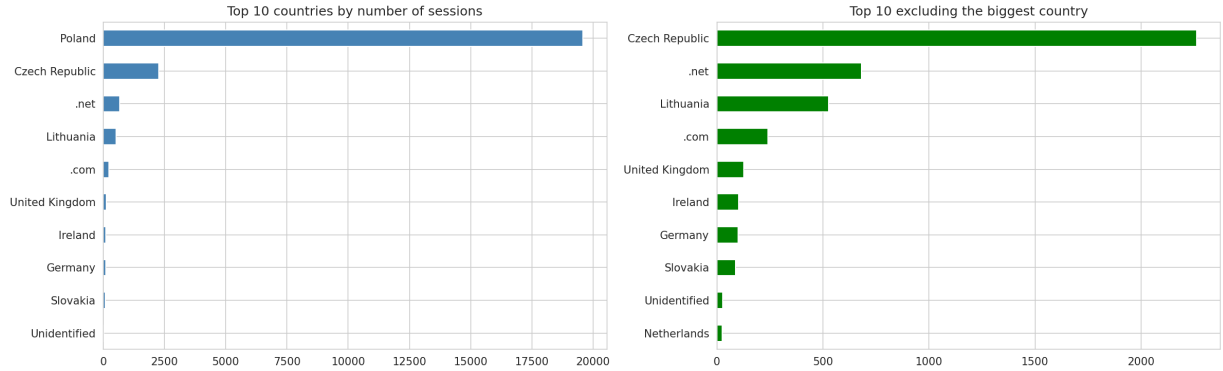


Figure 4: Top 10 countries by number of sessions (left) and the same without the biggest country (right).

What this shows: The shop depends on one market: **Poland gives 81.5% of the sessions** (19,582) and 81.0% of the clicks. The second country is the **Czech Republic with 2,261 sessions (9.4%)**, then the .net domain (681), Lithuania (527) and .com (240). All other countries have very small numbers, for example the United Kingdom has 127 sessions, Ireland 102 and Germany 101.

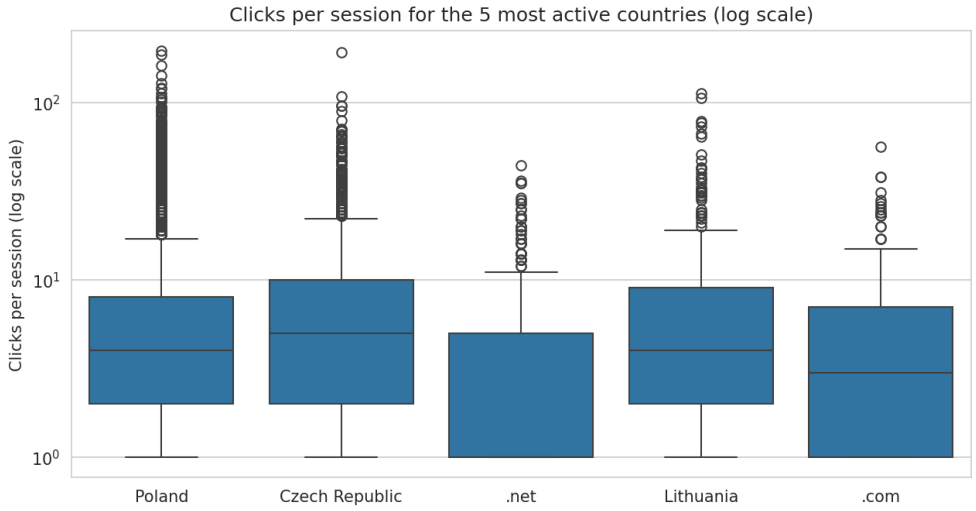


Figure 5: Clicks per session for the five most active countries (log scale).

What this shows: The typical session is similar in Poland (median 4 clicks), Lithuania (4) and the Czech Republic (5, average 7.96). The **.net group is different: its median is only 1 click**, and 53.3% of its sessions have a single click, compared with 20.1% in Poland. This is unusual and could mean automated or low-quality traffic (for example robots or ISP proxies).

4.4 Price and behaviour (Question 3)

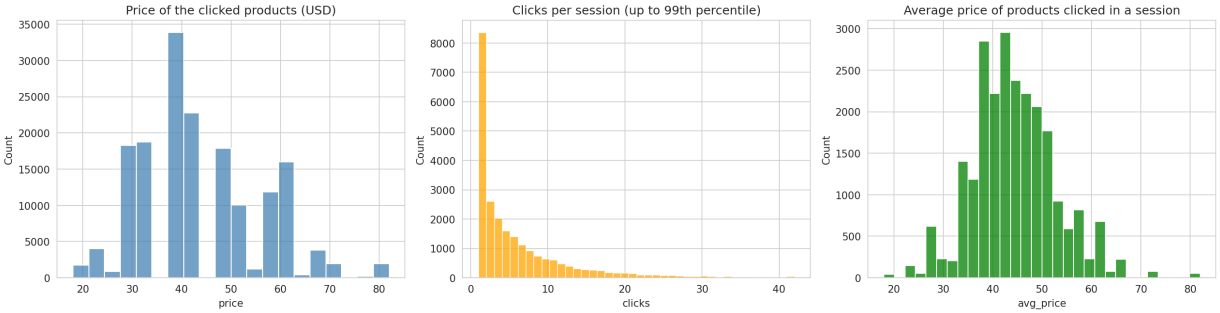


Figure 6: Price of clicked products (left), clicks per session (middle, up to the 99th percentile) and average price of products in a session (right).

What this shows: Prices go from USD 18 to USD 82, and the median is USD 43. Most clicks are on products priced USD 33 to USD 48 (the most common price is USD 38). Clicks per session are strongly right-skewed: most sessions are short and a few are very long. Average session price looks like a bell shape around USD 44.

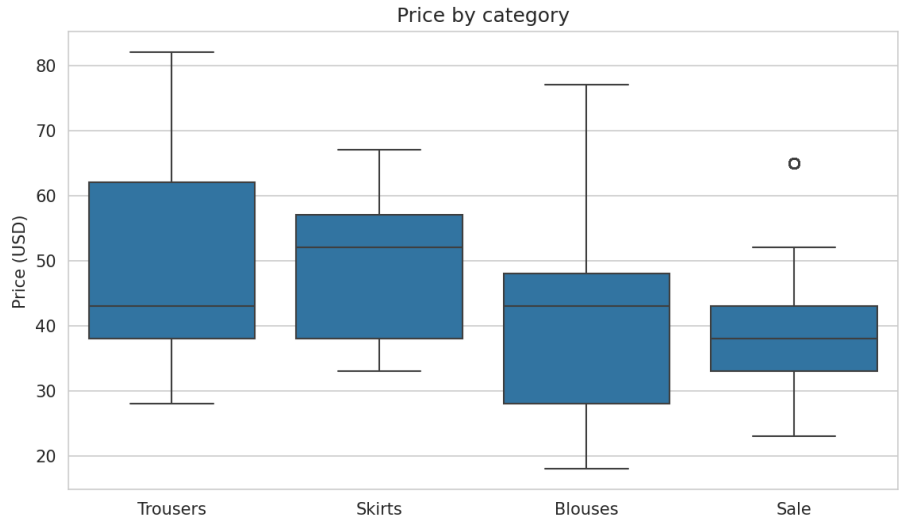


Figure 7: Price by category.

What this shows: Skirts have the highest median price (USD 52), trousers and blouses are in the middle (USD 43) and sale products are the cheapest (USD 38). Trousers have the widest price range.

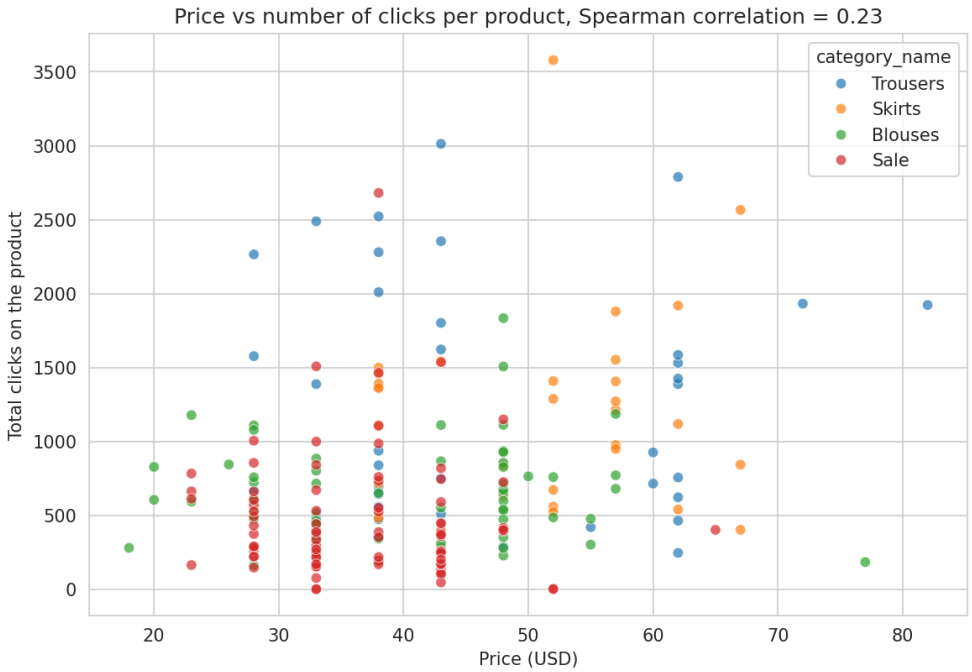


Figure 8: Total clicks per product versus price (each point is one product).

Price band (USD)	Products	Average clicks per product
Up to 30	39	636
31 - 40	75	702
41 - 50	59	702
51 - 60	25	960
Over 60	19	1,191

What this shows: Surprisingly, **more expensive products get more clicks per product, not fewer**: the correlation between price and clicks is weakly positive (Spearman 0.23), and products over USD 60 get about 1,191 clicks each against 636 for products up to USD 30. Products above their category average price get 799 clicks each against 728 for the others, and both groups together are 51% and 49% of all clicks. I should be careful with the reason: expensive products are mainly trousers and skirts, which are shown on the first pages and in good positions, so the effect of price cannot be separated from the effect of position. At least it shows that a higher price does not stop people from looking.

4.5 Popular products and colours

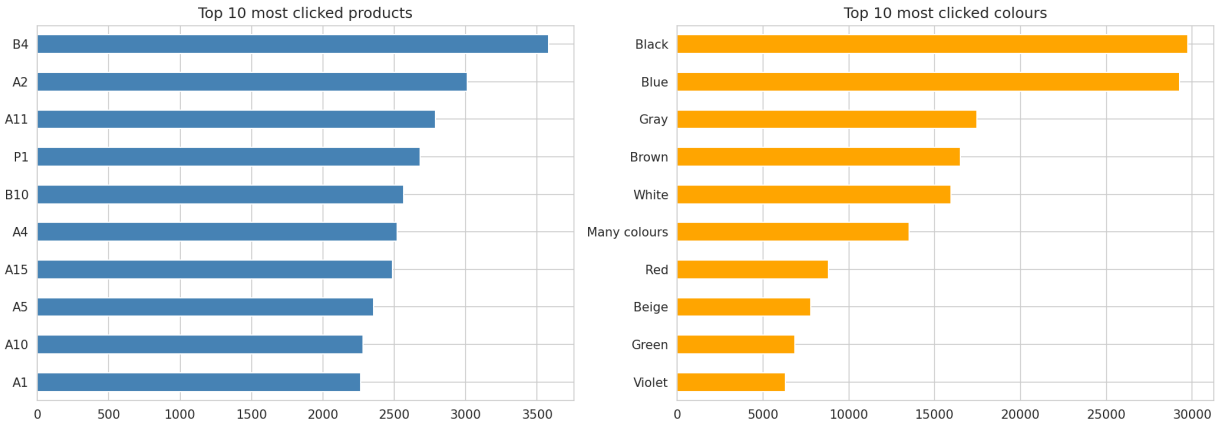


Figure 9: Top 10 most clicked products (left) and colours (right).

What this shows: The most clicked products are **B4 (3,579 clicks, a skirt), A2 (3,013), A11 (2,789), P1 (2,681) and B10 (2,566)**. The most clicked colours are **black (29,764) and blue (29,259)**, followed by gray (17,476), brown (16,516) and white (15,939). Neutral colours clearly dominate. Also, 74% of the clicks are on photos where the model is shown from the front and 26% on profile photos.

4.6 Behaviour inside a session

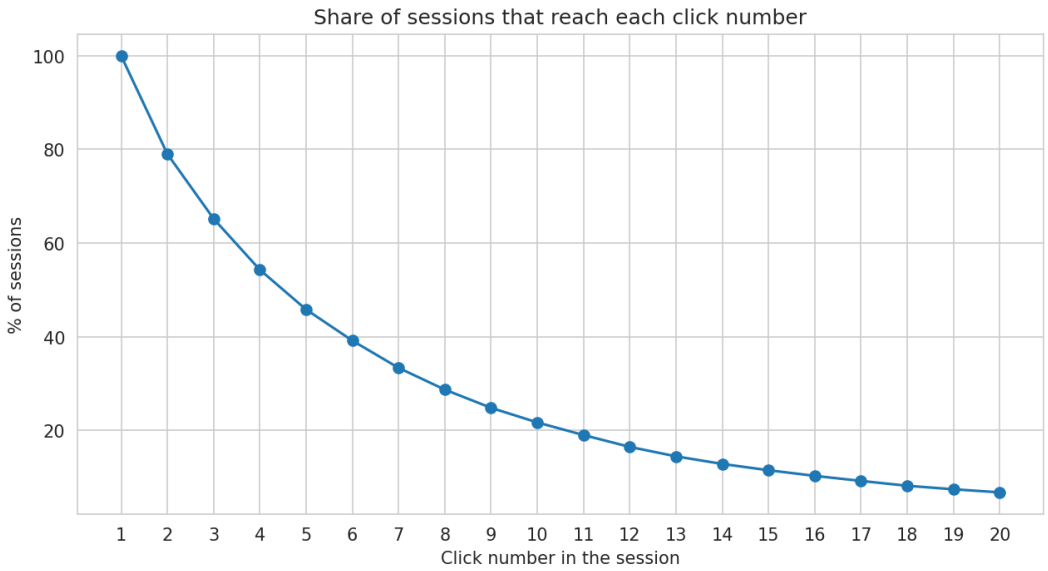


Figure 10: Share of sessions that reach each click number.

What this shows: Users leave quickly. All sessions have a first click, but only **79.0%** reach a second click, 65.2% reach a third, **45.8% reach the fifth click, 21.7% the tenth** and only 6.8% reach the

twentieth. Overall, 21.0% of the sessions have only one click and 39.6% never leave page 1.

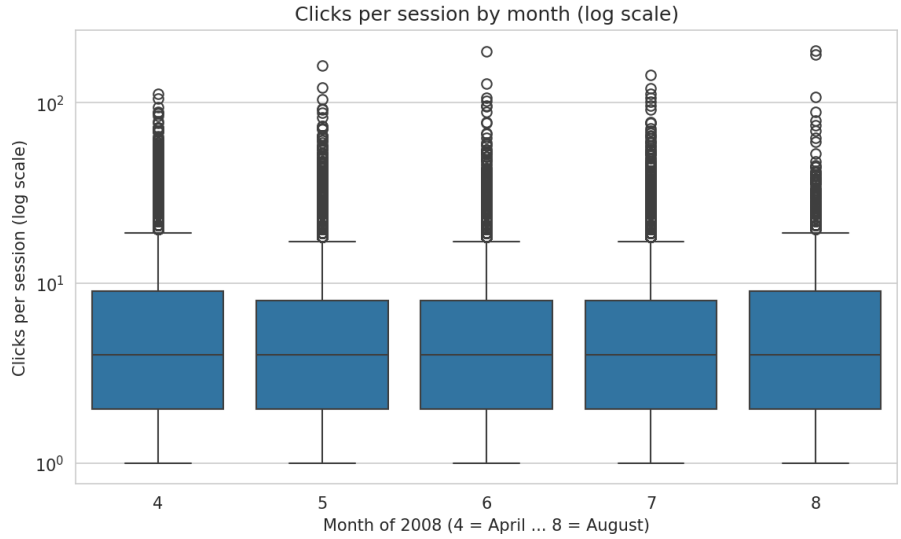


Figure 11: Clicks per session by month (log scale).

What this shows: The distribution of clicks per session is almost identical every month (median 4 clicks in all five months), so the way users browse did not change during the period.

4.7 Activity over time

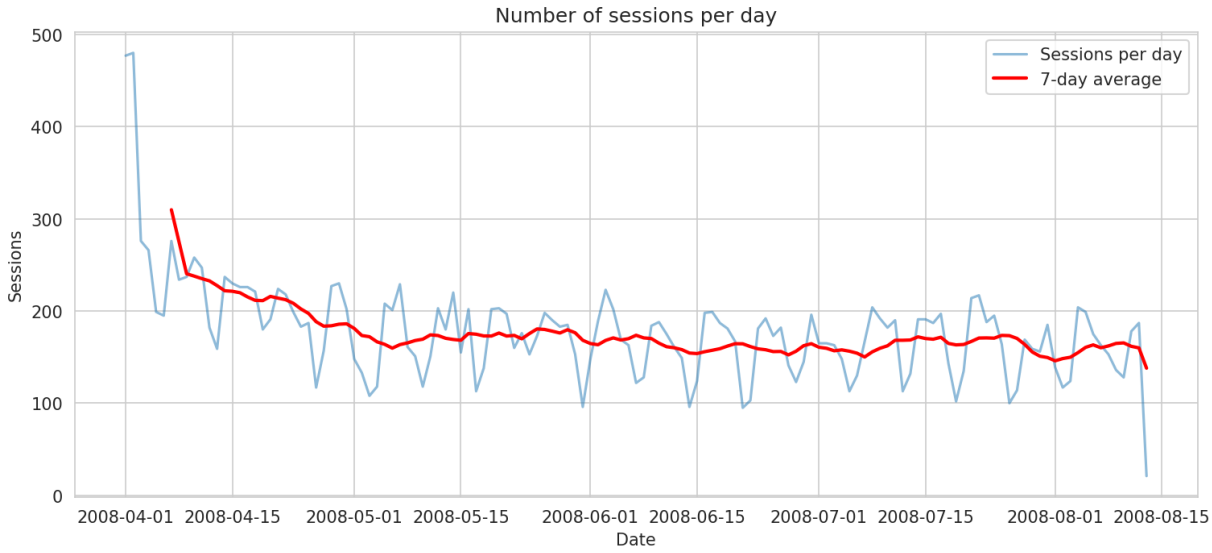


Figure 12: Number of sessions per day with a 7-day average.

What this shows: The shop had about 178 sessions per day on average. Activity was highest at the very start of the data (477 sessions on 1 April and 480 on 2 April), then fell and stayed fairly stable at about 170 sessions a day from mid-April on. Monthly sessions were 6,941 (April), 5,206 (May), 4,884 (June), 5,071 (July) and 1,924 (August, which is incomplete because the data stops on 13 August, so the drop at the end is not real).

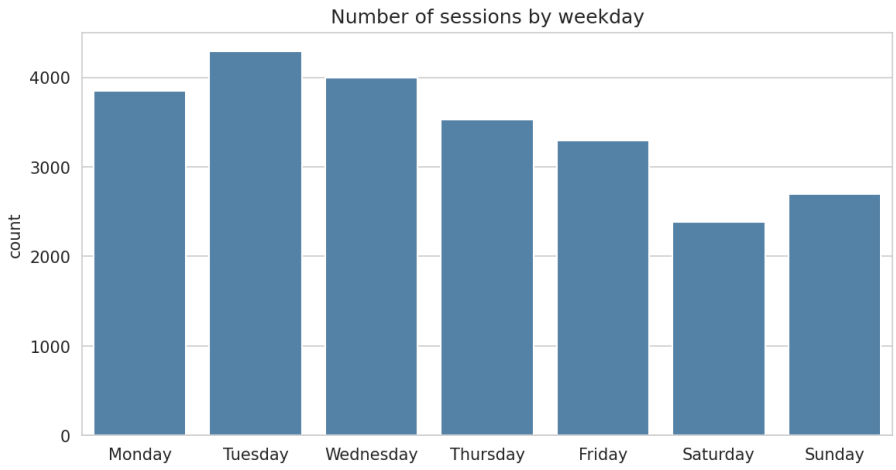


Figure 13: Number of sessions by weekday.

What this shows: Tuesday is the busiest day (4,287 sessions), followed by Wednesday (3,994) and Monday (3,846). **Saturday is the quietest** (2,383) and Sunday is also low (2,693). So the shop is used more on working days than on weekends.

4.8 Engagement as a proxy for conversion (Questions 7 and 8)

A session is 'highly engaged' if it has more than 8 clicks (more than 75% of all sessions). This is **24.8%** of the sessions. Remember that this is only a proxy, because the data does not say if the user bought something.

	Not engaged	Highly engaged
Average price of clicked products (USD)	44.63	43.60
% of clicks on above-average-price products	51.9%	51.0%
% of clicks in the sale category	17.3%	24.8%
Average number of categories visited	1.53	2.82

What this shows: Engaged sessions do **not** look different in price: they click on products of the same price level. The difference is that they **explore more categories (2.82 against 1.53) and look more at the sale category** (24.8% of their clicks against 17.3%). So engaged users compare across the shop and look for bargains.

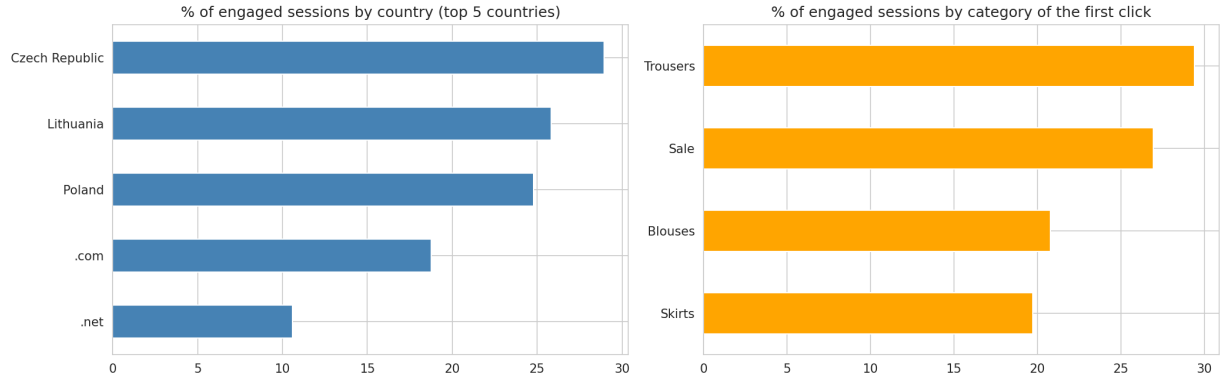


Figure 14: Share of highly engaged sessions by country (left) and by category of the first click (right).

What this shows: The **Czech Republic has the highest share of engaged sessions (28.9%)**, higher than Lithuania (25.8%) and Poland (24.8%), while .com (18.8%) and especially .net (10.6%)

are low. Sessions that start with **trousers (29.4%) or sale (27.0%)** are more often engaged than sessions that start with blouses (20.8%) or skirts (19.7%). The share of engaged sessions is almost the same in all months (23.6% to 26.9%) and on all weekdays (23.7% to 26.6%), so time of the week or year is not an important factor here.

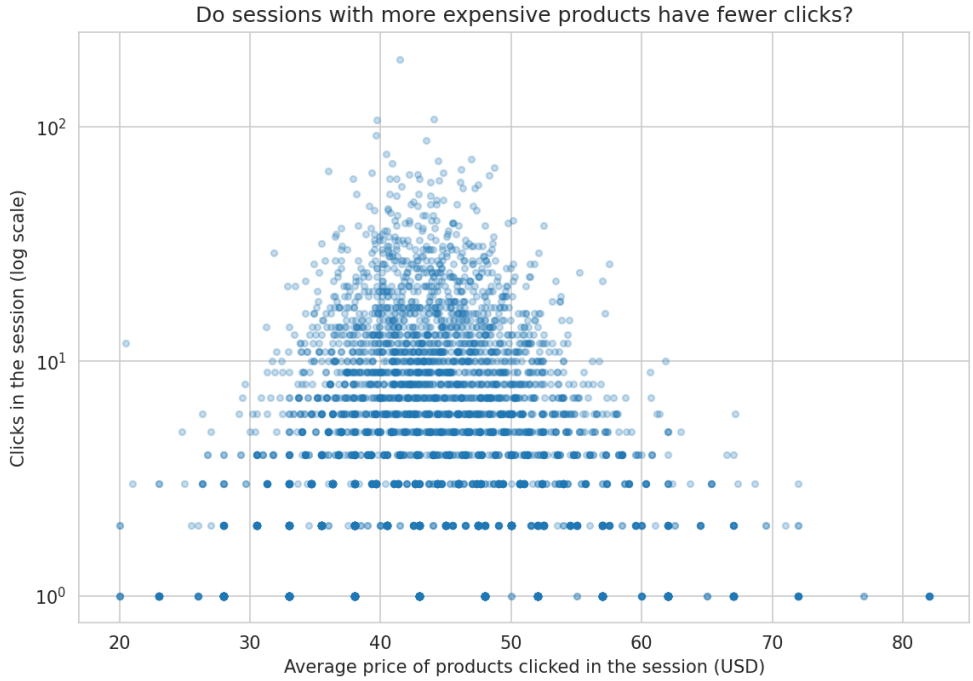


Figure 15: Average price of products in a session versus clicks in the session (sample of 5,000 sessions, log scale).

What this shows: There is practically no relationship between the average price of a session and its number of clicks (correlation -0.01). Users with cheap and with expensive tastes browse for the same number of clicks.

4.9 Correlation analysis

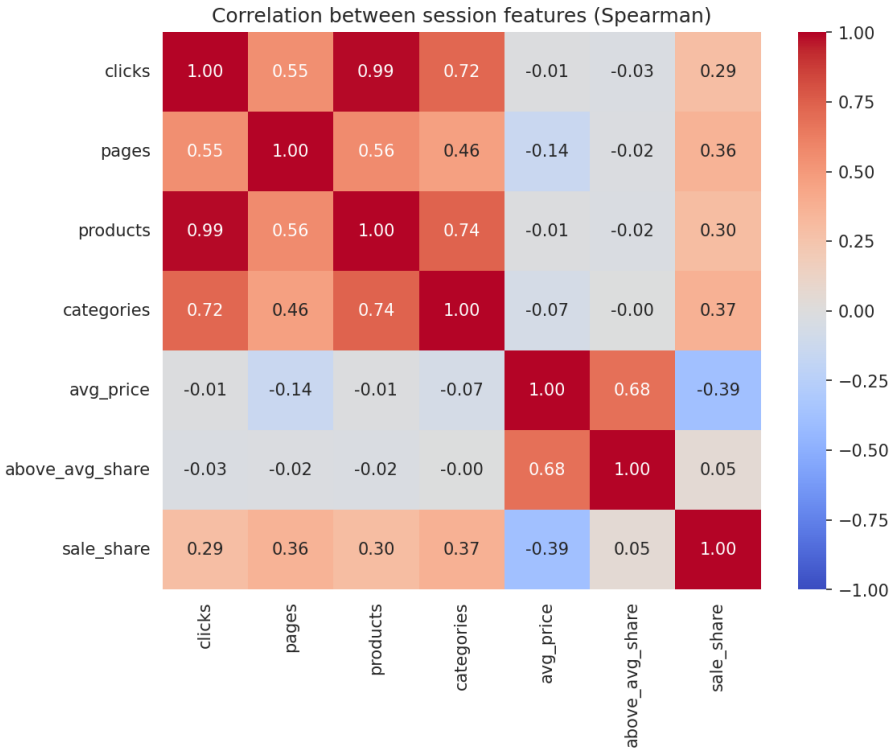


Figure 16: Spearman correlation between session features.

What this shows: Clicks and products are almost the same thing (0.99), because a session with more clicks views more different products. Clicks are also related to the number of categories visited (0.72) and the deepest page reached (0.55). These are real behaviour findings: users who click more also move across categories and pages. The sale share has a moderate relationship with clicks (0.29) and a negative one with average price (-0.39), because sale products are cheaper. Average price and the share of above-average-price clicks are related (0.68) as expected. Average price has almost no relationship with the number of clicks (-0.01), as seen in the scatter plot.

5. Answers to the Nine Questions

#	Question	Answer
1	Categories with most attention	Trousers: 49,742 clicks and 54% of sessions. Skirts and trousers get the most clicks per product (about 1,160); sale gets the least (473).
2	Pages with most clicks	Page 1 gets 56.5% of clicks, page 2 24.8%, page 3 11.7%, page 4 5.4%, page 5 1.7%.
3	Does price affect behaviour?	No sign that a higher price reduces clicks: a weak positive correlation (0.23) between price and clicks per product, but this is mixed with the effect of position. Session average price and session clicks are unrelated (-0.01).
4	Countries with most activity	Poland 81.5% of sessions, Czech Republic 9.4%, .net 2.8%, Lithuania 2.2%, .com 1.0%.
5	Most visited page locations	Top left (20.9%) and top middle (20.2%); the right side is weakest (13.1% and 12.5%).
6	Differences between categories	They differ in number of products (33 to 82), price (sale cheapest at USD 36, skirts dearest at USD 51), clicks per product and how deep in the shop they are shown.
7	Browsing vs purchasing patterns	No purchase data. Using the engagement proxy: engaged users (24.8% of sessions) visit about 2.8 categories against 1.5, and look more at the sale category.
8	Variables linked to conversion (proxy)	Number of categories visited, share of sale clicks, country (Czech Republic high, .net low) and the first category (trousers, sale high). Price, month and weekday show little link.
9	Unusual behaviour / outliers	1,962 sessions above 17 clicks, 17 sessions above 100 clicks (maximum 195); .net sessions have a median of 1 click and 53% single-click sessions.

6. Business Recommendations

- **Put the products you most want to sell on page 1 and in the top-left and top-middle positions.** Page 1 gets 56.5% of the clicks and page 5 only 1.7%, and the top row left and middle get about 41% of the clicks. The company should test (A/B test) whether these positions really cause more clicks.
- **Reduce the early drop-off.** 21% of sessions end after one click and only 46% reach the fifth click. The first page should help users go on, for example with related products, filters and an easy way to the next pages.
- **Recommend products across categories.** More than half of the sessions (52.9%) visit at least two categories and engaged users visit about 2.8. Suggestions like 'complete the look' (skirt with blouse) can make use of this behaviour.
- **Improve the sale section.** Sale has the most products (82) but the least attention per product (473 clicks). Show a smaller selection of the best sale items higher up, because users who start with sale are among the most engaged (27.0%).
- **Look at the Czech Republic and Lithuania for growth.** Poland is 81.5% of the sessions, but the Czech Republic has the most engaged users (28.9%) and Lithuania also does well (25.8%). Local-language content, prices and delivery offers for these markets could be tested.

- **Check the quality of the .net traffic.** More than half of its sessions are one click only. It should be filtered in the web analytics, or checked for robots, so that it does not spoil the statistics.
- **Do not be afraid of higher prices.** Expensive items are not clicked less. The company can promote premium trousers and skirts (which already get a lot of attention) and stock popular neutral colours (black, blue, gray).
- **Plan campaigns for working days.** Tuesday to Thursday have the most sessions and Saturday the fewest, so emails and offers should be sent early in the week.
- **Record purchases and add-to-cart events.** Without them the real conversion rate cannot be measured, and all findings about 'conversion' stay only a proxy.

7. Limitations

- The data has **no purchase information**, so conversion could not be measured. 'Highly engaged session' is only a proxy based on the number of clicks, and a user with many clicks may still not buy.
- The data covers only April to August 2008, and August is incomplete, so seasonality cannot be studied.
- Each row is a click on a product and the data comes from one clothing shop, so the results may not apply to other online shops. The data is also from 2008.
- About 81% of the traffic is from Poland, so results for other countries are based on much smaller numbers.
- Position on the page and category are mixed together (for example expensive items are mostly on early pages), so the effect of price on clicks cannot be separated.
- This is exploratory analysis. Correlations show relationships, not causes.

8. Conclusion

The clickstream shows a shop whose traffic comes mostly from Poland, where most users look at only a few products and leave early: half of the sessions end before the fifth click and more than half of all clicks are on the first page. Trousers and skirts get the most attention per product, the top-left and top-middle photo positions are clicked the most, and higher prices do not reduce clicks. Users who explore several categories and the sale section are the most engaged. The company can use these findings to improve product placement, make the first page more helpful, recommend products across categories, target the Czech and Lithuanian markets, and start recording purchases so that real conversion can be measured.

Resume description

Analyzed 165K+ clickstream records (24K shopping sessions) from an online clothing store using Python, Pandas, NumPy, Matplotlib and Seaborn to identify popular categories, page and country activity, session drop-off and engagement patterns, and recommended improvements to product placement and marketing.

Reference

Lapczynski M., Bialowas S. (2013) Discovering Patterns of Users' Behaviour in an E-shop - Comparison of Consumer Buying Behaviours in Poland and Other European Countries, Studia Ekonomiczne, nr 151, p. 144-153 (source of the dataset).